Chapter 7. The Bootstrap

Sampling distributions, bootstrap estimates of standard error and bias, resampling schemes

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Overview

(Part II of the book, "The bootstrap and modern nonparametric statistics," begins with this chapter.)

The **bootstrap** is the standard resampling method for estimating the sampling distribution of a statistic. With it one obtains confidence intervals for the parameter the statistic estimates, the bias and standard error of the estimator, and tests about the parameter: a very general method. The bootstrap is a nonparametric tool that makes statistical inference possible for any population. Even under parametric model assumptions it has the advantage of replacing mathematical-statistical calculation by computation.

Briefly, the bootstrap approximates the population by suitable resampling from the given sample. Compared with the other standard resampling method, the permutation test: the permutation test generates the distribution under the null by suitable shuffles, while bootstrap resampling is a method for inference about the estimation of parameters rather than for tests of hypotheses about them. Two examples show when the bootstrap can be used.

Example 7.1

The coefficient of variation of a population (standard deviation divided by the mean) is often used as an index of inequality. The scholarship amounts $x_1, \ldots, x_n$ of $n = 100$ randomly chosen Seoul National University students were surveyed and the estimate $\widehat{cv} = s/\bar{x}_n$ of the coefficient of variation computed. Is this estimator $\widehat{cv}$ biased compared with the population coefficient of variation cv? How large is $\mathrm{Var}(\widehat{cv})$?

Example 7.2

Kendall's tau $\tau = 2P\{(X - X')(Y - Y') > 0\} - 1$, the population measure of association of two variables (X, Y), can be estimated by Kendall's statistic τ_n, and the null distribution of τ_n under $H_0 : \tau = 0$ can be approximated by shuffles. If $\tau \neq 0$ appears to hold, can the distribution of τ_n be known without knowing the value of τ? If we want a confidence interval for τ, what method can be used?

In Example 7.1, with population mean and variance μ, σ^2, the population coefficient of variation is $cv = \sigma/\mu$, and we are interested in the properties of its estimator $\widehat{cv} = s/\bar{X}_n$. Among the properties of an estimator, the bias $E(\widehat{cv}) - cv$ and the standard error $\mathrm{se}(\widehat{cv}) := \sqrt{\mathrm{Var}(\widehat{cv})}$ are good measures of its uncertainty. The mean $E(\widehat{cv})$ and variance $\mathrm{Var}(\widehat{cv})$ can in principle be found by mathematical-statistical calculation under a parametric model, but their form is very complicated or (for normal populations[1]) has not yet been computed theoretically. Remember also that this bias and standard error depend on the distribution of the population. The bootstrap estimates the bias and standard error of a statistic from whatever population.

Kendall's statistic τ_n in Example 7.2 has a sampling distribution depending only on the parameter τ. Whatever the actual distribution, populations with the same value of τ give the same sampling distribution of τ_n, but τ is unknown, so the sampling distribution of τ_n is hard to specify. Even then, the bootstrap gives the sampling distribution of τ_n and confidence intervals for τ.

[1] Hendricks and Robey (1936) computed the approximate density of $\widehat{cv}$ for $X_i \sim N(\mu, \sigma^2)$, but its form is very complicated. In this paper of almost 100 years ago, to verify or test whether the computed density is correct, 10240 coin tosses were (actually) carried out; to approximate a normal variable, the number of heads in 10 coin tosses was used, approximately normal. By comparison, today one can draw random samples directly from the normal distribution, and repetitions in the millions rather than thousands are easy.

7.1 Sampling distributions and standard errors of statistics

The sampling distribution and the standard error of a statistic are essential for quantifying its uncertainty. From a population with distribution F, compute from n independent observations $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} F$ a statistic $\hat{\theta}_n = T(\mathbb{X}_n) = T(X_1, \ldots, X_n)$, regarded as an estimator of a parameter $\theta (= \theta(F)$, a value depending on $F)$. The sampling distribution of $\hat{\theta}_n$ is its distribution under the given population, and its standard error is the standard deviation of that distribution. The sampling distribution or standard error answers (probabilistically) how close the estimator $\hat{\theta}_n$ is to θ.

First, sampling distributions and standard errors in a hypothetical situation. The data `flights2.RData` record the scheduled departure and actual departure times of all passenger flights departing from all US airports in 2015, giving the departure delay. In 2015 there were 5,819,079 flights in all; regard them as the population. (It is a finite population, but with very many units, so treat it as infinite.) Figure 7.1 of the book visualizes 5000 of the five million records chosen at random.

A delay below 0 means the flight departed earlier than scheduled. If what we want to know from these data is the population median θ, we have the whole population, so we know $\theta = -2$ (minutes). Indeed more than half the flights departed earlier than scheduled. Note that in ordinary data analysis the parameter value is never known.

We have all flights, but suppose we could survey only $n = 100$ flights: the observed data are 100 delays $X_1, \ldots, X_n$ chosen at random from the five million of the population. How good an estimator of the population median θ is the sample median $\hat{\theta}_n = \text{median}\{X_1, \ldots, X_n\}$?

Since the population is accessible, the effect of the randomness of the sample on the sample median can be checked by simulation. Drawing a random sample of n from the population and computing its sample median $\hat{\theta}_n$ gives one observation of the sample median. If we sample again, how different will this observation be? Since the population is known, we can sample many times and compute the sample medians $\hat{\theta}_n^{(1)}, \hat{\theta}_n^{(2)}, \hat{\theta}_n^{(3)}, \ldots$ of each. If we could repeat this infinitely and see the distribution of $\{\hat{\theta}_n^{(i)} : i = 1, 2, 3, \ldots, \infty\}$, that is exactly the distribution of the sample median computed from n samples (Figure 7.2 of the book).

Without infinite repetition, repeating random sampling from the population sufficiently many times ($M = 100000$) and computing $\hat{\theta}_n$ approximates its sampling distribution (Figure 7.3 of the book, a histogram). If we observed a single sample, its sample median would be one value of this sampling distribution. (The sampling distribution shows that the sample median is very likely to be -2 or -1.) In the same way the standard error $\text{se}(\hat{\theta}_n)$ can be approximated: with $\hat{\theta}_n^{(i)}$ the median of each sample and $\bar{\hat{\theta}}_n^{(\cdot)}$ their mean,

$$\text{se}(\hat{\theta}_n) \approx \widehat{\text{se}}_M(\hat{\theta}_n) = \sqrt{\frac{1}{M-1}\sum_{i=1}^{M}\left(\hat{\theta}_n^{(i)} - \bar{\hat{\theta}}_n^{(\cdot)}\right)^2} = 0.759.$$

Now suppose the population is inaccessible, and we obtained only $n = 100$ observations with sample median $\hat{\theta}_n = \text{median}\{x_1, \ldots, x_n\} = -1$. (Compared with the sampling distribution, $\hat{\theta}_n = -1$ is a quite plausible value.) If the sampling distribution of $\hat{\theta}_n$ were approximately normal, the standard error $\text{se}(\hat{\theta}_n) = 0.759$ would give the 95% interval for θ

$$\hat{\theta}_n \pm 1.96\,\text{se}(\hat{\theta}_n) \approx [-2.49, 0.49].$$

In reality the sampling distribution is neither normal nor is the standard error $\text{se}(\hat{\theta}_n)$ known. Without access to the population, how can we know the sampling distribution of $\hat{\theta}_n$? And how to estimate the standard error $\text{se}(\hat{\theta}_n)$?[2] To answer this, the bootstrap regards the given sample as a (hypo-

[2]For some parametric models and statistics the sampling distribution is easy to obtain. E.g. if $X_i \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$ and the statistic is $T(\mathbb{X}_n) = \bar{X}_n$ (the sample mean), normal theory gives $\bar{X}_n \sim N(\mu, \sigma^2/n)$, so the sampling distribution is $N(\mu, \frac{\sigma^2}{n})$ and the standard error $\text{se}(\bar{X}_n) = \sigma/\sqrt{n}$. If σ is unknown, plugging in the sample standard deviation s gives an approximate standard error.

thetical) population and follows exactly the approximation process for the sampling distribution above.

7.2 Bootstrap samples and estimating sampling distributions

We saw that the sampling distribution of a statistic $T(\mathbb{X}_n)$ can be approximated by repeated sampling from the population. In practice the population is unavailable, so the bootstrap estimates the population distribution from the observations and performs repeated resampling with this estimated distribution as a hypothetical population: these are bootstrap samples.

With observations $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} F$ from a population whose form F cannot be specified, the best estimator of the population distribution function $F(x) = P(X \leq x)$ is the empirical distribution function

$$F_n(x) = \frac{1}{n} \sum_{i=1}^{n} I(X_i \leq x).$$

The larger the sample, the closer the empirical distribution function is to the actual one (the difference between $F(x)$ and $F_n(x)$ is about $1/\sqrt{n}$). The empirical distribution F_n is the discrete distribution with weight $\frac{1}{n}$ on each observation X_i, so if the X_i have no ties, F_n is the discrete uniform distribution on $\mathbb{X}_n = \{X_1, \ldots, X_n\}$. To draw n independent observations from this distribution F_n, sample with replacement from $\mathbb{X}_n$.

The bootstrap sampling method that approximates the sampling distribution of a statistic $\hat{\theta}_n = T(\mathbb{X}_n)$ from observations $\mathbb{X}_n = \{x_1, \ldots, x_n\}$ of one population:

Nonparametric bootstrap resampling:

1. Draw n observations from $\mathbb{X}_n = \{x_1, \ldots, x_n\}$ by simple sampling with replacement. These observations $\mathbb{X}_n^* = \{x_1^*, \ldots, x_n^*\}$ are called a bootstrap sample.
2. Compute the statistic $\hat{\theta}_n^* = T(\mathbb{X}_n^*)$ from the bootstrap sample.
3. Repeat steps 1–2 a predetermined number B of times, obtaining $\hat{\theta}_n^{*(b)}$ for $b = 1, \ldots, B$.
4. The empirical distribution of $\{\hat{\theta}_n^{*(b)} : b = 1, \ldots, B\}$ is the **bootstrap sampling distribution** of $\hat{\theta}_n$, an estimate of the sampling distribution of $\hat{\theta}_n$.

Figure 7.4 of the book shows this schematically. The values $\hat{\theta}_n^{*(b)}$ computed from bootstrap samples are also called bootstrap replicates.

The bootstrap sampling distribution has two errors: the error of approximating the population distribution $F(x)$ by $F_n(x)$ (of order $1/\sqrt{n}$), and the error of approximating the distribution of $\hat{\theta}_n^*$ by the B bootstrap replicates $\hat{\theta}_n^{*(b)}$ (of order $1/\sqrt{B}$). B can be made large enough, so the performance of the bootstrap is governed more by reducing the first error (i.e. increasing the sample size) than the second.

Example 7.3

Suppose we have data `dat` of $n = 10$ flight delays. Two bootstrap samples drawn at random in R:

```
> dat
 [1]  -5  12   9  35  -5 -10   1  -1  -3  -6
> (bootstrap.dat.1 = sample(dat,n, replace = TRUE))
 [1]  -3  -1 -10  12  -3  -1 -10   9 -10  -1
> (bootstrap.dat.2 = sample(dat,n, replace = TRUE))
 [1]  -6   1   9  -5   1 -10  -1  -6  35  -5
```

The original data `dat` contain only one observation -1, but in a bootstrap sample drawn with replacement this observation can be drawn several times. The median of the original data is $\hat{\theta}_n = -2$, and in the bootstrap samples $\hat{\theta}_n^{*(1)} = -2$, $\hat{\theta}_n^{*(2)} = -3$.

Example 7.4

A sample size $n = 10$ is too small, so take $n = 100$: repeating the bootstrap resampling $B = 2000$ times gives the empirical distribution of $\{\hat{\theta}_n^{*(b)} : b = 1, \ldots, B\}$, the bootstrap sampling distribution, shown in the book. Compare with the sampling distribution of Figure 7.3.

```
n = 100
dat = sample(delay, n) # delay: population of 5 million departure delays
B = 2000
med.boot = replicate(B, median( sample(dat,n, replace = TRUE)))
```

7.3 Bootstrap estimates of bias and standard error

Call the sampling distribution of a statistic $\hat{\theta}_n$ G (density g). (This sampling distribution is determined by the parameter θ, and more broadly by the population F and the sample size n.) Denote its estimate, the bootstrap sampling distribution, by G^* (density g^*). The properties of the statistic $\hat{\theta}_n$ relative to the target parameter θ are summarized by the bias

$$\text{bias}(\hat{\theta}_n) = E(\hat{\theta}_n) - \theta = \int \hat{\theta}_n \, dG - \theta = \int \hat{\theta}_n g(\hat{\theta}_n) \, d\hat{\theta}_n - \theta,$$

and the variance

$$\text{Var}(\hat{\theta}_n) = \int (\hat{\theta}_n - \mu_{\hat{\theta}_n})^2 \, dG = \int (\hat{\theta}_n - \mu_{\hat{\theta}_n})^2 g(\hat{\theta}_n) \, d\hat{\theta}_n, \qquad \mu_{\hat{\theta}_n} = E(\hat{\theta}_n),$$

or standard error $\text{se}(\hat{\theta}_n) = (\text{Var}(\hat{\theta}_n))^{1/2}$. To estimate the bias and standard error, substitute the bootstrap sampling distribution G^* for the unknown sampling distribution G in these formulas: these are the bootstrap estimates of bias and standard error.

First, the bootstrap estimate of the standard error:

$$\widehat{\text{se}}_{\text{Boot}}(\hat{\theta}_n) = \Big(\frac{1}{B-1} \sum_{i=1}^{B} (\hat{\theta}_n^{*(b)} - \mu_{\hat{\theta}_n^*})^2 \Big)^{1/2} \approx \Big(\int (\hat{\theta}_n^* - \mu_{\hat{\theta}_n^*})^2 \, dG^* \Big)^{1/2}, \tag{7.1}$$

where

$$\mu_{\hat{\theta}_n^*} = \frac{1}{B} \sum_{b=1}^{B} \hat{\theta}_n^{*(b)} = \int \hat{\theta}_n^* \, dG^*.$$

The first expression in (7.1) (the sample standard deviation) is the one mainly used.[3]

To compute the bootstrap estimate of the bias, we must replace not only the sampling distribution G but also the parameter θ. Just as G was replaced by G^*, the parameter $\theta = \theta(F)$, defined through the distribution of the population, can be replaced by $\hat{\theta}_n = \theta(F_n)$, defined on $\mathbb{X}_n$ (the hypothetical population). So the bootstrap estimate of the bias is

$$\widehat{\text{bias}}_{\text{Boot}}(\hat{\theta}_n) = \frac{1}{B} \sum_{b=1}^{B} \hat{\theta}_n^{*(b)} - \hat{\theta}_n = \mu_{\hat{\theta}_n^*} - \hat{\theta}_n.$$

[3] The sample variance of the bootstrapped statistics is used because the bootstrap resamples only B times, not infinitely often, and dividing by $B - 1$ reduces the bias (of order $1/B$).

Example 7.5

From the sample of $n = 100$ delays of Example 7.4, the bootstrap estimates of the standard error and bias of the median $\hat{\theta}_n$:

```
> sd(med.boot)
[1] 0.8294599
> (bias = mean(med.boot) - median(dat))
[1] 0.2405
```

Compared with the actual standard error and bias of the median, $\text{se}(\hat{\theta}_n) \approx 0.73$ and $\text{bias}(\hat{\theta}_n) \approx 0.48$,[a] each estimate still varies. With more than $n = 100$ observations the estimates become more accurate.

[a] The sample median is in general a biased estimator. In the delay example the population distribution is not continuous but a heavily right-skewed discrete distribution, so the bias of the sample median $\hat{\theta}_n$ is not 0. (The sample median $\hat{\theta}_n$ satisfies $\text{median}\{\hat{\theta}_n\} - \theta = 0$, i.e. it is median-unbiased.)

7.4 Parametric bootstrap resampling

If a specific parametric model can be assumed for the population from which the observations were sampled, using this fact increases accuracy. E.g. if the observations are success/failure or presence/absence, the population can only be a Bernoulli distribution with success probability p, so there is no way not to assume this parametric model. Or if the population is believed approximately normal, one can assume it is in the normal family $\{N(\mu, \sigma^2) : -\infty < \mu < \infty,\ \sigma^2 > 0\}$ and bootstrap.

A new data set for illustration:

Example 7.6

The March wind speeds (m/s) in Seoul, when spring cold spells hit, observed over ten years, are summarized below (Yoo and Kim, 2012). The data were originally continuous values but are recorded as discrete wind speeds (all speeds between 0 and 1 recorded as 0.5).

wind speed (m/s)	0.5	1.5	2.5	3.5	4.5	5.5	6.5	7.5	8.5	9.5
frequency	6421	14246	11953	7463	2322	659	120	11	3	0

A small wind generator is to be installed in Seoul, and its mean power output θ in March estimated. At wind speed v the output is known to be approximately $p(v) = 1 - 2v + v^2 - 0.05v^3$. What are the estimate of θ and its standard error?

With the wind speed a random variable $V \sim F$ (F the population of March wind speeds in Seoul), the parameter is

$$\theta = E(p(V)) = E(1 - 2V + V^2 - 0.05V^3),$$

and with observations v_i $(i = 1, \ldots, n)$ the natural estimator of θ is the moment estimator

$$\hat{\theta}_n = \frac{1}{n}\sum_{i=1}^{n} p(v_i) = 1.827462. \tag{7.2}$$

(Here n is not the number of speed categories, 10, but the total of all frequencies, $n = 43198$.) The estimator is a mean, so its standard error can be estimated by the formula

$$\text{se}(\hat{\theta}_n) = \Big(\frac{1}{n}\,\text{Var}\{p(V)\}\Big)^{1/2} \approx \Big(\frac{1}{n(n-1)}\sum_{i=1}^{n}\{p(v_i) - \overline{p(v.)}\}^2\Big)^{1/2} = \widehat{\text{se}}(\hat{\theta}_n). \tag{7.3}$$

Of course the standard error can also be estimated by nonparametric bootstrap resampling.

Wind speed is known to be modeled well by the Weibull distribution, so the population can be modeled by the Weibull family $\mathcal{W}$. The Weibull family $\mathcal{W} = \{f(\cdot; k, \lambda) : k > 0, \lambda > 0\}$ consists of the Weibull distributions $W(k, \lambda)$ with density

$$f(v; k, \lambda) = \frac{k}{\lambda}\Big(\frac{v}{\lambda}\Big)^{k-1} e^{-(v/\lambda)^k}, \qquad v \geq 0.$$

The mean and variance (i.e. bias and standard error) of the estimator $\hat{\theta}_n$ (7.2) can then be computed mathematically, but at a great cost of time and effort. In that case, or when mathematical calculation is impossible, the parametric bootstrap is used.

From the data one obtains the maximum likelihood estimators $(\hat{k}, \hat{\lambda})$ of the two Weibull parameters,[4] and with them $W(\hat{k}, \hat{\lambda})$, a better estimate of the population distribution than the empirical distribution function. Samples can be drawn freely from this parametric distribution $W(\hat{k}, \hat{\lambda})$: these are parametric bootstrap samples.

Parametric bootstrap resampling:

1. Assume the observations follow a parametric distribution $f(\cdot; \eta)$, where η is a parameter θ or a vector of parameters. Obtain from the data $\mathbb{X}_n$ the estimate $\hat{\eta}$ of the parameter η of this distribution. The statistic of interest is $\hat{\theta}_n = T(\mathbb{X}_n)$.
2. Draw n random observations from $f(\cdot; \hat{\eta})$. These observations $\mathbb{X}_n^* = \{x_1^*, \ldots, x_n^*\}$ are called a (parametric) bootstrap sample.
3. Compute the statistic $\hat{\theta}_n^* = T(\mathbb{X}_n^*)$ from the bootstrap sample.
4. Repeat steps 2–3 a predetermined number B of times, obtaining $\hat{\theta}_n^{*(b)}$ for $b = 1, \ldots, B$.
5. The empirical distribution of $\{\hat{\theta}_n^{*(b)} : b = 1, \ldots, B\}$ is the bootstrap sampling distribution of $\hat{\theta}_n$, an estimate of its sampling distribution.

When the parametric assumption holds, parametric bootstrap resampling approximates the unknown population more accurately and so estimates the standard error of a statistic more accurately. Whether parametric or nonparametric bootstrap resampling is done, the results are the bootstrap replicates $\{\hat{\theta}_n^{*(b)} : b = 1, \ldots, B\}$, so whichever resampling is used, the estimation of the standard error, and the confidence interval methods introduced later, are the same.

Example 7.7

For the data of Example 7.6, the maximum likelihood estimates of the two Weibull parameters are $(\hat{k}, \hat{\lambda}) = (1.94, 2.49)$, and the estimated parametric distribution $W(\hat{k}, \hat{\lambda})$ fits the data very well (figure in the book). The R code below carries out parametric bootstrap resampling for the moment estimator $\hat{\theta}_n$. With the observations in a vector `v` of length $n = 43198$, first estimate the two Weibull parameters, then resample bootstrap samples from $W(\hat{k}, \hat{\lambda})$.

```
> mle=weibull.mle(v, threshold = 0, interval = c(0,3))
> k.hat = mle$shape
> lambda.hat = mle$scale
>
> # theta.hat
> theta.hat = mean(1 - 2 * v + v^2 - 0.05 * v^3)
>
> set.seed(1)
> theta.boot = replicate(B,{
+   v.boot = rweibull(n, k.hat, lambda.hat)
+   pv.boot = 1 - 2 * v.boot + v.boot^2 - 0.05 * v.boot^3
```

[4] The maximum likelihood estimators of the two Weibull parameters have no closed-form solution, so they are found by numerical optimization; here `weibull.mle()` of R's `weibullness` package was used.

```
+   mean(pv.boot)
+ })
> head(theta.boot)
[1] 1.814683 1.842158 1.825138 1.830239 1.799006 1.799625
>
> se.boot.par = sd(theta.boot)
> bias.boot.par = mean(theta.boot) - theta.hat
```

The table compares (a) the standard error of $\hat{\theta}_n$ by formula (7.3), (b) the standard error and bias of $\hat{\theta}_n$ from the nonparametric bootstrap, and (c) those from the parametric bootstrap.

	formula (7.3)	nonparametric bootstrap	parametric bootstrap
standard error estimate	0.0122	0.0123	0.0125
bias estimate		0.0001	−0.009

The three standard error estimates for $\hat{\theta}_n$ are very similar. The estimator is a mean, hence theoretically unbiased, and the bootstrap bias estimates are also very close to 0.

When the parametric assumption is true, the parametric bootstrap gives more accurate estimates than the nonparametric bootstrap. In comparison, the nonparametric bootstrap can be used over a very wide range of populations.

7.5 Various bootstrap resampling schemes

The principle of bootstrap resampling is to reproduce the generating process of the original data as faithfully as possible. So far the observations from one population were assumed to be a simple random sample, so bootstrap samples were also obtained by simple random sampling from the empirical distribution F_n or an estimated parametric distribution $f(\cdot;\hat{\eta})$. Applying this principle to various situations gives bootstrap resampling suited to each. Two examples follow.

7.5.1 Bootstrap resampling when comparing several populations

Suppose the observations from k populations are

$$\begin{aligned} X_{11},\ldots,X_{1n_1} &\overset{\text{i.i.d.}}{\sim} F_1,\\ X_{21},\ldots,X_{2n_2} &\overset{\text{i.i.d.}}{\sim} F_2,\\ &\vdots\\ X_{k1},\ldots,X_{kn_k} &\overset{\text{i.i.d.}}{\sim} F_k. \end{aligned} \tag{7.4}$$

When the statistic $T(\mathbb{X}_n)$ computed from the sample $\mathbb{X}_n = \{X_{11},\ldots,X_{1n_1},\ldots,X_{k1},\ldots,X_{kn_k}\}$ is of interest, the bootstrap approximates the sampling distribution and variance of $T(\mathbb{X}_n)$. Here bootstrap resampling is done separately within each population. In nonparametric bootstrap resampling, the k different populations of (7.4) are replaced by the empirical distributions $\hat{F}_j$ of the samples from each, and bootstrap samples are drawn. Step 1 of resampling is:

1. Draw n_1 observations from the first population's sample $\{X_{11},\ldots,X_{1n_1}\}$ by simple sampling with replacement, $x^*_{11},\ldots,x^*_{1n_1}$. Likewise for each $j=2,\ldots,k$, draw n_j observations from the jth population's sample $\{X_{j1},\ldots,X_{jn_j}\}$ by sampling with replacement, $x^*_{j1},\ldots,x^*_{jn_j}$. The resampled observations $\{x^*_{11},\ldots,x^*_{1n_1},\ldots,x^*_{k1},\ldots,x^*_{kn_k}\}$ are the bootstrap sample.

The remaining steps 2–4 proceed exactly as steps 2–4 of nonparametric bootstrap resampling to obtain the bootstrap sampling distribution of $T(\mathbb{X}_n)$.

For parametric bootstrap resampling, each population F_j is expressed by a parametric model. E.g. under normality with equal variances, $F_j = N(\mu_j, \sigma^2)$, and with the maximum likelihood estimates of the parameters $\mu_1, \ldots, \mu_k, \sigma^2$ we obtain $\hat{F}_j = N(\hat{\mu}_j, \hat{\sigma}^2)$. Replace the k different populations F_j of (7.4) by $\hat{F}_j$, and for $j = 1, \ldots, k$ draw n_j new observations at random from $\hat{F}_j$ as the bootstrap sample. Then proceed with steps 3–5 of parametric bootstrap resampling as before to obtain the bootstrap sampling distribution of $T(\mathbb{X}_n)$.

Example 7.8

The acute leukemia gene expression data of Example 4.1 in Chapter 4 recorded gene expression scores of $n_1 = 47$ patients sampled from population ALL and $n_2 = 25$ from population AML. We estimate the standard error $\text{se}(D)$ of the statistic $D = \bar{X}_1 - \bar{X}_2$ by bootstrap resampling. With the observations in vectors `ALL` and `AML` of lengths 47 and 25, nonparametric bootstrap resampling:

```
md.boot = replicate(B,
          mean(ALL[sample(1:n1, n1, replace = TRUE)]) -
          mean(AML[sample(1:n2, n2, replace = TRUE)])
          )
```

The observed statistic is $D = -67.28$ and the bootstrap estimate of the standard error is `sd(md.boot)` $= 28.3$. The bootstrap sampling distribution of this statistic is shown as a histogram in the book.

7.5.2 Bootstrap resampling for bivariate data

The generating process of bivariate data $\{(X_1, Y_1), \ldots, (X_n, Y_n)\}$ can be viewed in roughly one of two ways. The first, as introduced in Chapter 5, views them as data sampled at random from a population F following a bivariate distribution:

$$(X_1, Y_1), \ldots, (X_n, Y_n) \overset{\text{i.i.d.}}{\sim} F. \tag{7.5}$$

The second, a model often assumed for inference in simple regression, views the "X" variable of the two as given in advance or fixed, and only the "Y" variable as random. Under model (7.5) this can be seen as the conditional distribution of Y_i after fixing the observations X_i, but consider the special case of the linear regression model

$$Y_i = \alpha + \beta x_i + \epsilon_i, \qquad \epsilon_i \overset{\text{i.i.d.}}{\sim} F_\epsilon. \tag{7.6}$$

More complex regression models can be assumed, e.g. a regression curve $\alpha + \beta x_i + \gamma x_i^2$ instead of the line $\alpha + \beta x_i$, or errors ϵ_i whose distribution varies with x_i instead of identically distributed ϵ_i; here assume the simplest simple regression model (7.6) is true.

Both models (7.5) and (7.6) generate bivariate data $\{(X_1, Y_1), \ldots, (X_n, Y_n)\}$, but in resampling, the bootstrap sampling must differ according to which model is assumed to have generated the data. The nonparametric bootstrap resampling for each data-generating model:

- **Bootstrapping pairs (cases):** for data (7.5) sampled at random from a population following a bivariate distribution, with the sample written as vectors $\mathbf{x}_i = (X_i, Y_i)$, $\mathbf{x}_1, \ldots, \mathbf{x}_n$, draw n vectors with replacement from them to get the bootstrap sample $\{\mathbf{x}_1^*, \ldots, \mathbf{x}_n^*\}$ and do bootstrap resampling.
- **Residual bootstrap:** for data (7.6) where the "X" variable is given in advance and the "Y" variables do not have the same distribution, the ϵ_i, which have the same distribution, are

resampled. The errors ϵ_i are not observable, so the residuals of the fitted regression line,

$$e_i = Y_i - \hat{\alpha} - \hat{\beta}x_i, \qquad i = 1, \ldots, n,$$

are resampled instead. I.e. draw n residuals with replacement from $\{e_1, \ldots, e_n\}$ to get bootstrap residuals $\{e_1^*, \ldots, e_n^*\}$, and collect the bootstrap-resampled

$$\begin{pmatrix} Y_i^* \\ x_i^* \end{pmatrix} = \begin{pmatrix} \hat{\alpha} + \hat{\beta}x_i + e_i^* \\ x_i \end{pmatrix}, \qquad i = 1, \ldots, n,$$

to generate the bootstrap sample $\{(x_1^*, Y_1^*), \ldots, (x_n^*, Y_n^*)\}$. (In every bootstrap sample $x_i^* = x_i$ is fixed.)

The remaining steps 2–4 of the nonparametric bootstrap are carried out as is to obtain the bootstrap sampling distribution of the statistic of interest.

Which data-generating model fits better in which case? Both models leave the shapes of the distributions F and F_ϵ unrestricted, so both can be thought of as nonparametric models. But model (7.6) also has a "parametric model" $E(Y_i) = \alpha + \beta x_i$ besides the nonparametric distribution of ϵ_i, so it is called a semi-parametric model. The residual bootstrap, assuming model (7.6), may be inaccurate when the parametric assumption (simple linearity) does not hold. By comparison, the pairs bootstrap, assuming the nonparametric model (7.5) without specific parameters, can be used in most cases.

Recall that inference on the slope β in regression always assumes the model $E(Y_i) = \alpha + \beta x_i$ is true. Also, most inference in regression assumes the explanatory "X" variable is fixed, i.e. model (7.6). So for the bootstrap sampling distribution and standard error of the regression slope estimator $\hat{\beta}$, the residual bootstrap is the standard. Nonparametric inference in regression is explained in Chapter 10.

For bivariate or multivariate statistics in contexts other than regression, assume model (7.5) and use the pairs bootstrap.

Example 7.9

With the GDP and happiness data of the World Happiness Report seen in Chapter 5 (Example 5.2), infer Kendall's tau τ between the two variables. For these data Kendall's statistic is $\tau_n = 0.503$, and the permutation test of $H_0 : \tau = 0$ against $H_1 : \tau > 0$ gives $p < 0.01$: $\tau > 0$ very significantly (R code below). The permutation distribution of τ_n used in the permutation test is shown in Figure 7.5 of the book; it is the distribution when the null $\tau = 0$ is true.

What, then, is the sampling distribution and standard error of τ_n when $\tau \neq 0$? To answer, compute the bootstrap sampling distribution and estimate the standard error by bootstrap resampling of pairs of observations. In the code below for the pairs bootstrap, note that the ids of the observations are resampled, so the same observations are drawn for the two variables `gdp` and `score`.

```
> wh = read.csv("data/whappiness.csv")
> score = wh$Score; gdp = wh$GDP.per.capita
> (k.obs = cor(gdp,score, method = "k"))
[1] 0.5032363
> B = 1999
> k.boot = replicate(B, {
+   id.boot = sample(1:n,n, replace = TRUE)
+   cor(gdp[id.boot], score[id.boot], method = "k")
+ })
> sd(k.boot)
[1] 0.2096958
```

The bootstrap estimate of the standard error is 0.21. The distribution of the bootstrap replicates τ_n^* (`k.boot`) is the bootstrap sampling distribution, an estimate of the sampling distribution of τ_n in the population (estimated to have $\tau \neq 0$); see Figure 7.5 of the book. The random permutation distribution from shuffles, contrasted with bootstrap resampling, was generated with this R code:

```
> # random permutation test
> M = 1999
> n = length(gdp)
> k.perm = replicate(M, {
+   id.perm = sample(1:n, n, replace = FALSE)
+   cor(gdp,score[id.perm], method = "k")
+ })
> (pvalue.perm = mean(c(k.obs,k.perm) >= k.obs))
[1] 0.0075
```

Instead of sampling with replacement (`replace = TRUE`) as in the bootstrap, the permutation test samples without replacement (`replace = FALSE`). The bootstrap draws the pairs of observations (x_i, y_i) together, preserving the correlation between the two variables, while the permutation test keeps x_i fixed and shuffles only y_i, (deliberately) destroying any correlation between the two variables.